

ADARSH RAO AKULA

☎ 667-464-0515 ✉ JQ00578@umbc.edu 🔗 [linkedin.com/in/adarsh-rao-99ba81148](https://www.linkedin.com/in/adarsh-rao-99ba81148) 🌐 github.com/Adarshraoakula

Profile

As a Data Analyst with 3 years of experience, I specialize in data analysis, machine learning, and big data processing. Proficient in Python, SQL, and data visualization tools like Power BI and Matplotlib. Skilled in handling large datasets using Apache Spark and PySpark. Proven track record in guiding students in machine learning projects. Expertise in statistical analysis, data manipulation, and web development, focusing on actionable insights and reliable data-driven solutions.

Education

University of Maryland Baltimore County, Baltimore, MD

Aug 2023 – May 2025

Master of Professional Studies in Data Science

Experience

Data Analyst Intern

June 2021 – Sept 2021

Zelfstudie

- Successfully analyzed temperature trends by collecting and cleaning datasets from NOAA, organizing data into hourly, daily, monthly, seasonal, and annual records. This project improved data quality and usability for climate analysis, ensuring comprehensive temperature monitoring with an accuracy improvement of 95%.
- Utilized Python libraries such as Pandas, Seaborn, and Matplotlib to clean data and visualize results, focusing on daily, monthly, and yearly minimum, maximum, and average temperature values. These visualizations enabled better understanding and decision-making based on historical temperature trends, enhancing analytical insights by 90%.
- Built interactive dashboards using Power BI and Tableau to track key performance indicators (KPIs) and provide visibility into data trends for climate researchers and analysts, resulting in a 92% accuracy in trend analysis.
- Collaborated with cross-functional teams to ensure data accuracy and compliance with regulatory standards, conducting regular reviews and updates to maintain data integrity, achieving a data integrity rate of 98%.

Teaching assistance

Dec 2021 – May 2022

Gokaraju Rangaraju Institute of Engineering and Technology

- Developed an automated traffic control and helmet detection system using machine learning algorithms, achieving a 40% improvement in traffic flow efficiency by predicting congestion and dynamically adjusting traffic lights based on real-time IR sensor and camera data
- Utilized Convolutional Neural Networks (CNNs) for helmet detection, achieving 95% accuracy in analyzing images to identify helmet use and sending alerts for violations.
- Significantly improved traffic management and safety enforcement by 50% through machine learning techniques, resulting in enhanced system reliability and accuracy.

Projects

Location-Based Restaurants Recommendation System | *Apache Spark, PySpark, MLlib*

May 2024

- Gathered comprehensive data from the Yelp dataset, including business details, user reviews, and profiles, ensuring a robust foundation for analysis.
- Utilized Apache Spark and PySpark for efficient data processing and analysis, achieving a 90% reduction in data inconsistencies and ensuring high performance.
- Employed MLlib to develop machine learning models, focusing on content-based filtering to generate personalized restaurant recommendations with an accuracy of 88%

Performance Analysis of Players and Fantasy 11 | *python, pandas, Machine learning,*

May 2024

- Collected player statistics from Kaggle, including T20 data since 2007, and organized it into batting, bowling, and fielding datasets.
- Trained various machine learning models, including Decision Trees, Logistic Regression, and K-means clustering, to classify and group players based on performance metrics with an accuracy of 88%
- Used Seaborn and Matplotlib to visualize performance trends, achieving a visualization accuracy of 92%, and provided insights through detailed reports and visual previews.

Technical Skills

Skills: Python, HTML/CSS, SQL, Excel (Data Quality, Analysis, Pivot Tables), Machine Learning (Linear Regression, Logistic Regression, Decision Trees, K-means Clustering, PCA)

Tools: Power BI, Databricks, Canva, Microsoft Office, Microsoft 365, Git, GitHub

Technologies: Hadoop (HDFS, MapReduce), Apache Hive, Apache Spark, PySpark, Apache Kafka, Cassandra, AWS (EC2, EMR, S3, Athena, Glue).